

advantage of Lightrun is its dynamic logging nature, allowing configuration of the time interval, and line numbers to insert log and log level (debug, info, error, to name a few). This reduces the actual logging size stored in the disk as only the required logs generated for debugging are kept.

Importance of logging and monitoring of running containers

Observability services, like logging and telemetry monitoring, use various metrics from infrastructure and platform services to troubleshoot any non-functional issues like performance, scalability and right-sizing requirements.

When working with Kubernetes applications (e.g., AKS in Azure, EKS in AWS or GKE in Google Cloud), we can enable an integrated native logging service with Lightrun monitoring services. It requires a Lightrun agent to be enabled to collect various metrics (as per configuration) and send it to the telemetry store for dashboard visualisation and other observability requirements.

Observability is based on three kinds of data.

- **Information:** This helps us collect vital statistics about the platform's health, alerts, failures, etc.
- **Inference:** This is conceived from the information gathered.
- **Insights:** Based on these predictions, action is taken for recovering the platform in case of failures or enable proactive action like vertical scaling of services.

Operational activities in cloud platforms like Azure Kubernetes Service (AKS) are typically divided as control

plane and data plane. Control plane is the driver, which can be used to create and manage any cloud resource. Data plane is the operational activity, which handles the capabilities of the resources created and managed by the control plane.

As an example, if you create a database, a storage account, a virtual machine or an app service, it is called a control plane. The actual operation activity like database operations, read/write activities on a storage account, and accessing virtual machines or application deployment on the app service are all functionalities of the data plane.

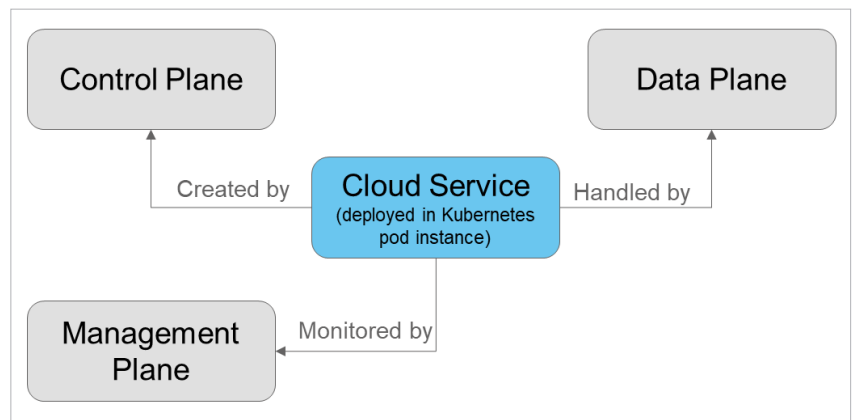


Figure 1: Anatomy of Kubernetes pod instance

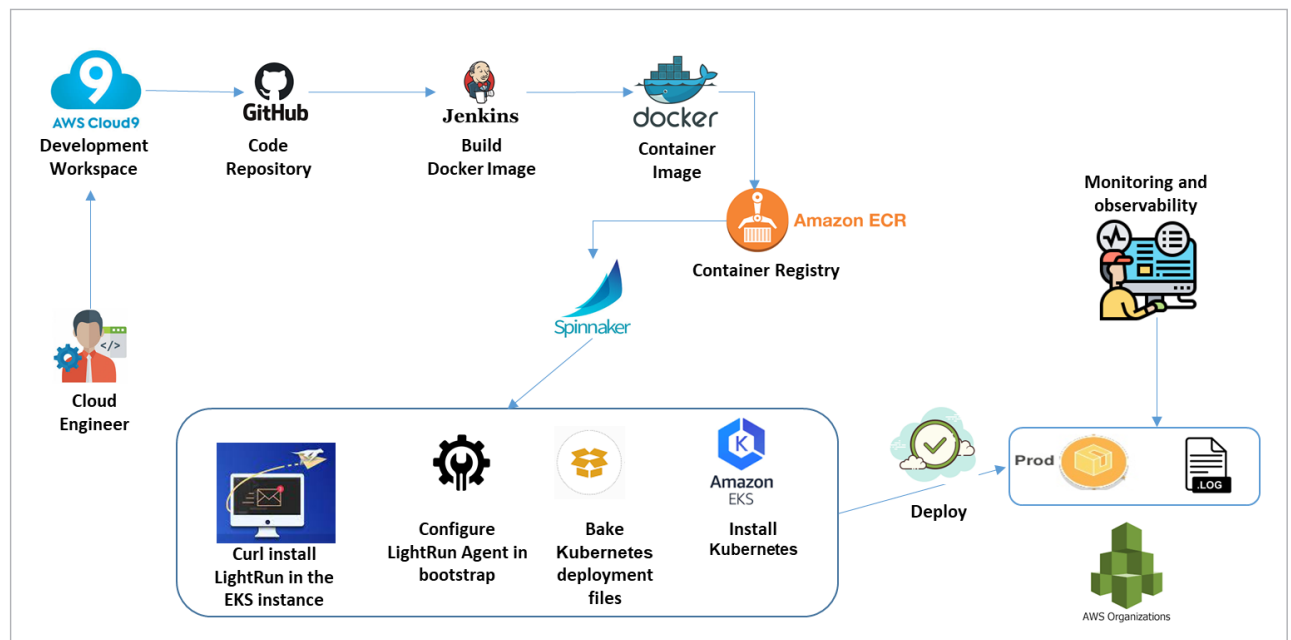


Figure 2: Deploying an example EKS instance with Lightrun instrumentation for log monitoring